

Lecture Note 7: Large Sample Properties of MLE

14.380: Statistical Methods in Economics, Fall 2026

1 Maximum Likelihood Estimation

Let $f(\cdot|\theta)$ with $\theta \in \Theta$ be a parametric family. Let $X = (X_1, \dots, X_n)$ be a random sample from distribution $f_1(\cdot|\theta_0)$ with $\theta_0 \in \Theta$. Then the joint pdf is $f(x|\theta) = \prod_{i=1}^n f_1(x_i|\theta)$ where $x = (x_1, \dots, x_n)$. The log-likelihood is $\ell(\theta|x) = \sum_{i=1}^n \log f_1(x_i|\theta)$. The maximum likelihood estimator is, by definition,

$$\hat{\theta}_{ML} = \arg \max_{\theta \in \Theta} \ell(\theta|x).$$

The FOC is

$$\frac{1}{n} \sum_{i=1}^n \frac{\partial \ell_1(\hat{\theta}_{ML}|x_i)}{\partial \theta} = 0.$$

1.1 Consistency and Asymptotic Normality

Thus, the MLE can be thought of as the method of moments estimator corresponding to the first information equality. So we can expect that the MLE is consistent. Indeed, the theorem below gives the consistency result for MLE:

Theorem 1 (MLE consistency). *In the setting above, assume that (1) θ_0 is identifiable, i.e. for any $\theta \neq \theta_0$, there exists x such that $f(x|\theta) \neq f(x|\theta_0)$, (2) the support of $f(\cdot|\theta)$ does not depend on θ , and (3) θ_0 is an interior point of parameter space Θ . Then $\hat{\theta}_{ML} \rightarrow_p \theta_0$.*

The proof of MLE consistency will be given in 14.382 and 14.385. What the proof does is show that the function $g(\theta) = \mathbb{E}_{\theta_0} \ell_1(\theta|X_i)$ (here $X_i \sim f_1(x_i|\theta_0)$) is maximized at $\theta = \theta_0$ and the random process $\frac{1}{n} \ell(\theta|X)$ converges to the function $g(\theta)$ in a uniform manner in probability. Then it argues that the maximizer $\hat{\theta}_{ML}$ of the process will converge to θ_0 .

Once we know that the estimator is consistent, we can think about the asymptotic distribution of the estimator. The next theorem gives the asymptotic distribution of MLE:

Theorem 2 (MLE asymptotic normality). *In the setting above, assume that conditions (1)-(3) in the MLE consistency theorem hold. In addition, assume that (4) $f_1(x_i|\theta)$ is thrice differentiable with respect to θ and we can interchange integration with respect to x and differentiation with respect to θ , and (5) $|\partial^3 \log f_1(x_i|\theta)/\partial \theta^3| \leq M(x)$ and $\mathbb{E}[M(X_i)] < \infty$. Then,*

$$\sqrt{n}(\hat{\theta}_{ML} - \theta_0) \xrightarrow{d} N(0, I_1^{-1}(\theta_0))$$

Proof. This is a sketch of the proof as it misses an important step. By definition, $\frac{\partial \ell(\hat{\theta}_{ML}|x)}{\partial \theta} = 0$. By the Taylor theorem with a remainder, there is some random variable $\tilde{\theta}$ with value between θ_0 and $\hat{\theta}_{ML}$ such that

$$\frac{\partial \ell(\hat{\theta}_{ML}|X)}{\partial \theta} = \frac{\partial \ell(\theta_0|X)}{\partial \theta} + \frac{\partial^2 \ell(\tilde{\theta}|X)}{\partial \theta^2} (\hat{\theta}_{ML} - \theta_0).$$

So,

$$\sqrt{n}(\hat{\theta}_{ML} - \theta_0) = \frac{-\frac{1}{\sqrt{n}} \frac{\partial \ell(\theta_0|X)}{\partial \theta}}{\frac{1}{n} \frac{\partial^2 \ell(\tilde{\theta}|X)}{\partial \theta^2}}.$$

Since $\hat{\theta}_{ML} \rightarrow_p \theta_0$ and $\tilde{\theta}$ is between θ_0 and $\hat{\theta}_{ML}$, $\tilde{\theta} \rightarrow_p \theta_0$ as well. From $\tilde{\theta} \rightarrow_p \theta_0$, one can prove that

$$\frac{1}{n} \frac{\partial^2 \ell(\tilde{\theta}|X)}{\partial \theta^2} - \frac{1}{n} \frac{\partial^2 \ell(\theta_0|X)}{\partial \theta^2} = o_p(1).$$

We will not discuss this result here since it requires knowledge of the concept of asymptotic equicontinuity which we do not cover in this class. You will learn it in 14.385. Note, however, that this result does not follow from the Continuous mapping theorem since we have a sequence of random functions $\ell(\theta|X)$ instead of just one non-random function. Suppose we believe in this result. Then, by the Law of large numbers,

$$\frac{1}{n} \frac{\partial^2 \ell(\theta_0|X)}{\partial \theta^2} = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \log f_1(X_i|\theta_0)}{\partial \theta^2} \rightarrow_p \mathbb{E} \left[\frac{\partial^2 \log f_1(X_i|\theta_0)}{\partial \theta^2} \right] = -I_1(\theta_0).$$

Next, by the first information equality, $\mathbb{E} \left[\frac{\partial \log f_1(X_i|\theta_0)}{\partial \theta} \right] = 0$ while $Var \left[\frac{\partial \log f_1(X_i|\theta_0)}{\partial \theta} \right] = I_1(\theta_0)$. Thus, by the Central limit theorem,

$$\frac{1}{\sqrt{n}} \frac{\partial \ell(\theta_0|X)}{\partial \theta} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial \log f_1(X_i|\theta_0)}{\partial \theta} \xrightarrow{d} N(0, I_1(\theta_0)).$$

Finally, by the Slutsky theorem,

$$\sqrt{n}(\hat{\theta}_{ML} - \theta_0) \xrightarrow{d} N(0, I_1^{-1}(\theta_0)).$$

□

Remark 1 (Vector-valued parameters). The same result holds for vector-valued parameters. Suppose $\theta \in \mathbb{R}^k$, and let $s(\theta|X_i) = \partial \log f_1(X_i|\theta)/\partial \theta$ denote the score of one observation, now a $k \times 1$ vector. Under analogous regularity conditions,

$$\sqrt{n}(\hat{\theta}_{ML} - \theta_0) \xrightarrow{d} N(0, I_1(\theta_0)^{-1}),$$

where

$$I_1(\theta_0) = \mathbb{E} [s(\theta_0|X_i)s(\theta_0|X_i)'] = -\mathbb{E} \left[\frac{\partial^2 \ell_1(\theta_0|X_i)}{\partial \theta \partial \theta'} \right]$$

is the $k \times k$ Fisher information matrix. The information equalities and the plug-in estimators of the asymptotic variance discussed below extend in the same way, with squared scores replaced by outer products of the score vector and second derivatives replaced by the Hessian matrix.

One interpretation of MLE asymptotics is that the MLE is asymptotically efficient (it attains

the Rao-Cramer bound in very large samples).

Example 1. Let $X_1, \dots, X_n$ be a random sample from a distribution with pdf $f(x|\lambda) = \lambda \exp(-\lambda x)$. This distribution is called exponential. Its log-likelihood for one draw is $\ell_1(\lambda|x_i) = \log \lambda - \lambda x_i$. So $\frac{\partial \ell_1(\lambda|x_i)}{\partial \lambda} = 1/\lambda - x_i$ and $\frac{\partial^2 \ell_1(\lambda|x_i)}{\partial \lambda^2} = -1/\lambda^2$. So Fisher information is $I_1(\lambda) = 1/\lambda^2$. Let us find the MLE for λ . The log-likelihood for the whole sample is $\ell(\lambda|x) = n \log \lambda - \lambda \sum_{i=1}^n x_i$. The FOC is $\frac{n}{\hat{\lambda}_{ML}} - \sum_{i=1}^n x_i = 0$. So $\hat{\lambda}_{ML} = \frac{1}{\bar{X}_n}$. Its asymptotic distribution is given by $\sqrt{n}(\hat{\lambda}_{ML} - \lambda) \xrightarrow{d} N(0, \lambda^2)$.

Example 2. Maximum likelihood is the workhorse of *discrete choice* analysis, where economic structure delivers the parametric family. Suppose individual i chooses to work ($y_i = 1$) or not ($y_i = 0$), and models the decision as maximizing utility: the individual works if

$$U_i = x_i' \beta + \epsilon_i \geq 0,$$

where x_i collects observed characteristics (the wage, family structure, and so on) and ϵ_i is unobserved taste. If ϵ_i follows the logistic distribution with cdf $\Lambda(u) = \frac{e^u}{1+e^u}$, then

$$P(y_i = 1 | x_i) = P(\epsilon_i \geq -x_i' \beta) = \Lambda(x_i' \beta),$$

using the symmetry of the logistic distribution. This is the *logit model*. Treating the x_i as fixed, the log-likelihood of the sample is

$$\ell(\beta) = \sum_{i=1}^n \{y_i \ln \Lambda(x_i' \beta) + (1 - y_i) \ln (1 - \Lambda(x_i' \beta))\}.$$

Using $\Lambda'(u) = \Lambda(u)(1 - \Lambda(u))$, the score takes a strikingly simple form:

$$\frac{\partial \ell(\beta)}{\partial \beta} = \sum_{i=1}^n (y_i - \Lambda(x_i' \beta)) x_i,$$

so the MLE sets the prediction errors $y_i - \Lambda(x_i' \beta)$ orthogonal to the characteristics — the discrete-choice analogue of the least-squares normal equations. Differentiating once more,

$$\frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta'} = - \sum_{i=1}^n \Lambda(x_i' \beta) (1 - \Lambda(x_i' \beta)) x_i x_i',$$

which is negative semidefinite for *every* β : the log-likelihood is globally concave, so the MLE is easy to compute numerically and any solution of the first-order condition is the global maximizer. The Fisher information for a single observation is

$$I(\beta) = \mathbb{E} [\Lambda(X_i' \beta) (1 - \Lambda(X_i' \beta)) X_i X_i'],$$

and applying the asymptotic normality theorem for the MLE (in its vector-parameter form, per

the remark above),

$$\sqrt{n}(\hat{\beta}_{MLE} - \beta) \xrightarrow{d} N(0, I(\beta)^{-1}).$$

The logit model is the seed of a large literature: McFadden (1974) developed the random-utility framework for choices among many alternatives, and logit-based demand systems are the foundation of modern structural analysis of markets in industrial organization.

1.2 Inference using MLE

We will have a longer discussion about how to estimate the asymptotic variance of the MLE $(I_1(\theta_0))^{-1}$ later when we discuss asymptotic tests. Right now we introduce three possibilities.

First of all, if $I_1(\theta)$ is a continuous function in θ (which is needed for asymptotic results), then given that $\hat{\theta}_{ML}$ is consistent for θ_0 , the quantity $(I_1(\hat{\theta}_{ML}))^{-1}$ is consistent for $(I_1(\theta_0))^{-1}$.

Second, by definition of Fisher information, it equals the expectation of either the negative second derivative of the log-likelihood or the squared score. Instead of taking the expectation, one may approximate it by taking averages. For example,

$$\hat{I} = -\frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \ell_1(\hat{\theta}_{ML}|X_i)}{\partial \theta^2}$$

will be a consistent estimator of the Fisher information.

The third idea to be used in this context is the parametric bootstrap. Assume $\hat{\theta}_{ML}$ is the MLE we obtained from our sample of size n . For $b = 1, \dots, B$ do the following:

- Simulate sample $X_b^* = (X_{1b}^*, \dots, X_{nb}^*)$ as i.i.d. draws from $f_1(x_i|\hat{\theta}_{ML})$ (that is, assuming that $\hat{\theta}_{ML}$ is the true parameter value).
- Find the MLE using sample X_b^* ; denote it θ_b^* .

Calculate the sample variance of $(\theta_1^*, \dots, \theta_B^*)$; it gives the bootstrap approximation to $(nI_1(\theta_0))^{-1}$. You may also do bootstrap bias correction using a similar procedure.

1.3 When MLE Asymptotic Theory Fails Us...

Example 3. For asymptotic normality of MLE, we should have common support. Let us see what might happen otherwise. Let $X_1, \dots, X_n$ be a random sample from $U[0, \theta]$. Then, $\hat{\theta}_{ML} = \max_i X_i$. So $\sqrt{n}(\hat{\theta}_{ML} - \theta)$ is always nonpositive. So it does not converge to a mean-zero normal distribution. In fact, $\mathbb{E}[\max_i X_i] = (n/(n+1))\theta$ and $V(\max_i X_i) = \theta^2 n / ((n+1)^2(n+2)) \approx \theta^2/n^2$. On the other hand, if the theorem worked, we would have $V(\max_i X_i) \approx 1/(nI_1(\theta))$. The MLE happens to be “super-consistent” here, meaning that it converges to the true value at a faster speed than the regular parametric speed of $1/\sqrt{n}$.

Example 4. Now, let us consider what might happen if the true parameter value θ_0 were on the boundary of Θ . Let $X_1, \dots, X_n$ be a random sample from distribution $N(\mu, 1)$ with $\mu \geq 0$. As an exercise, check that $\hat{\mu}_{ML} = \bar{X}_n$ if $\bar{X}_n \geq 0$ and 0 otherwise. Suppose that $\mu_0 = 0$. Then $\sqrt{n}(\hat{\mu}_{ML} - \mu_0)$ is always nonnegative. So it does not converge to a mean-zero normal distribution.

Example 5. Finally, note that it is implicitly assumed in both the consistency and asymptotic normality theorems that parameter space Θ is fixed, i.e. independent of n . In particular, the number of parameters should not depend on n . Indeed, let

$$X_i = \begin{pmatrix} X_{1i} \\ X_{2i} \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_i \\ \mu_i \end{pmatrix}, \begin{pmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{pmatrix} \right)$$

for $i = 1, \dots, n$, and let $X_1, \dots, X_n$ be mutually independent. One can show that if the sample size (n) increases to infinity, the MLE for σ^2 is inconsistent in this case, though a consistent estimator for σ^2 exists.

Remark 2. What is interesting, though we will not show it here, is that the bootstrap does not help in these cases. That is, the bootstrap approximation to the distribution of $\hat{\theta}_{ML}$ is not close to the true finite-sample distribution of $\hat{\theta}_{ML}$.

1.4 Pseudo-MLE

Let us have a sample $X = (X_1, \dots, X_n)$ i.i.d. from some distribution. We do not know what distribution it is; let us assume it has pdf $g(x_i)$. But we wrongly assumed a specific parametric family, that is, we assumed $X_i \sim f_1(x_i|\theta)$. What happens if we do MLE? MLE will estimate a “pseudo-true” parameter value θ_0 which minimizes in some sense the distance between $g(\cdot)$ and the family $f_1(\cdot|\theta)$. In particular:

$$\theta_0 = \arg \max_{\theta} \int \log[f_1(x_i|\theta)]g(x_i)dx_i = \arg \max_{\theta} \mathbb{E}[\log f_1(X_i|\theta)].$$

Of course, we can equivalently rewrite this as

$$\begin{aligned} \theta_0 &= \arg \min_{\theta} -\mathbb{E}[\log f_1(X_i|\theta)] \\ &= \arg \min_{\theta} -\mathbb{E}[\log f_1(X_i|\theta)] + \mathbb{E}[\log g(X_i)] \\ &= \arg \min_{\theta} \mathbb{E}[\log(\frac{g(X_i)}{f_1(X_i|\theta)})] \end{aligned}$$

Importantly, $\mathbb{E}[\log(\frac{g(X_i)}{f_1(X_i|\theta)})] = KL(g(\cdot), f_1(\cdot|\theta))$ is known as the KL-divergence between $g(\cdot)$ and $f_1(\cdot|\theta)$. It is a directed distance on probability density functions; the larger $KL(g(\cdot), f_1(\cdot|\theta))$, the worse $f_1(\cdot|\theta)$ approximates $g(\cdot)$. So, we can equivalently interpret the maximum likelihood estimand as choosing the parametric distribution $f_1(\cdot|\theta_0)$ that best approximates the true pdf $g(\cdot)$. Of course, whether the parameter θ_0 is of interest depends on the setting.

Under some regularity conditions $\hat{\theta}_{ML} \rightarrow_p \theta_0$, and in most parts the logic of the proof of the theorem about normality will hold. However, the information equality would fail. Define

$$\Sigma_1 = \mathbb{E} \left[\left(\frac{\partial \log f_1(X_i|\theta_0)}{\partial \theta_0} \right)^2 \right],$$

$$\Sigma_2 = -\mathbb{E} \left[\frac{\partial^2 \log f_1(X_i|\theta_0)}{\partial \theta_0^2} \right],$$

where expectations in both cases are taken assuming that $X_i \sim g(\cdot)$. If g is not in the parametric family, then in general $\Sigma_1 \neq \Sigma_2$. But using the logic of the proof, we can prove that

$$\sqrt{n}(\hat{\theta}_{ML} - \theta_0) \xrightarrow{d} N(0, \Sigma_2^{-1} \Sigma_1 \Sigma_2^{-1})$$

This asymptotic variance $\Sigma_2^{-1} \Sigma_1 \Sigma_2^{-1}$ is often called White's variance, due to White's (1980) paper, and the associated standard errors are called White's standard errors.